Progress Report 2

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Project: Smart Door Anomaly Detection & Cybersecurity for Elderly Care

Course: CSIS 4495 – Applied Research Project, Section 3

Reporting Period: Feb 24 – March 2, 2026

Report Date: March 2, 2026

Work Date/Hours Logs

Date	Hours	Description
Feb 28, 2026	3.5	Debugged dummy face encoding bug (scripts/capture_faces.py, register_faces.py) - replaced np.random.rand(128) with HOG+Sobel extraction | Implemented real facial feature extraction
Mar 1, 2026	3	Expanded synthetic dataset from 30 to 60 days in data_generator.py | Added 5th anomaly type: failed_access
Mar 2, 2026	2.5	Testing, validation, repo check-in, report creation| Team meeting
Total	**9**	

Summary Description of Work Done

This week, I identified and fixed a critical bug in the facial recognition pipeline: both capture_faces.py and register_faces.py were registering faces using completely random 128-dimensional vectors (np.random.rand(128)), making genuine recognition impossible(possibility of a false detection). I replaced this with proper **HOG (Histogram of Oriented Gradients) + Sobel edge detection** feature extraction that reads actual face images from the sample folders (Advitya, Eric, Reubin) and extracts real biometric vectors.

Over the weekend, I also expanded the anomaly detection training dataset from **30 days to 60 days** with improved confidence score distributions. I introduced a new **5th anomaly type**

called "failed_access" — simulating repeated low-confidence recognition attempts (0.30–0.55 range) that represent suspicious behavior or potential spoofing attempts.

Issues Encountered:

- Face registration was using random data instead of actual facial features
- Dataset was too small for robust anomaly detection
- Face detection system never recognized correctly
- Limited anomaly diversity in synthetic data

Solutions:

- Migrated to real HOG+Sobel feature extraction
- Expanded temporal dataset coverage
- Added realistic spoofing/failed-access patterns

Repo Check In of Implementation Completed

Files checked in this period:

- **scripts/capture_faces.py** — Updated with HOG+Sobel feature extraction
- **scripts/register_faces.py** — Modified to use real facial encodings
- **data/data_generator.py** — Expanded dataset and added 5th anomaly type
- **data/synthetic_dataset.csv** — Regenerated with 60-day coverage

All changes committed with detailed messages explaining the fixes. The face registration pipeline now captures real facial biometrics and the anomaly training data is significantly more robust.

AI Use Section

AI Tool	Version / Account Type	Specific Use	Value Addition
GitHub Copilot	Free Tier	HOG+Sobel implementation templates	Reviewed suggestions for correctness; verified 128D output against existing API contracts

AI Tool	Version / Account Type	Specific Use	Value Addition
ChatGPT 5.3	Pro Tier	Time-series anomaly pattern design	Selected domain-specific threat patterns; manually adjusted distributions based on security knowledge

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Appendix: Prompt History

Prompt 1: "How do I extract HOG+Sobel features from face images in Python?"

- **Used:** Code template for feature extraction
- **Modified:** Added scikit-image Sobel preprocessing and ensured 128D output compatibility

Prompt 2: "How should I structure synthetic access logs with realistic security anomalies?"

- **Used:** Suggestion to create multiple anomaly types
- **Modified:** Created domain-specific patterns (brute_force, spoofing, failed_access) with realistic confidence distributions matching real threat models